Conserv Genet (2016) 17:1081–1091
DOI 10.1007/s10592-016-0845-2

RESEARCH ARTICLE

Geographic distribution of genetic diversity in populations of Rio Grande Chub *Gila pandora*

René Galindo[1] · Wade D. Wilson[2] · Colleen A. Caldwell[3]

Received: 11 July 2015 / Accepted: 23 April 2016 / Published online: 6 May 2016
© Springer Science+Business Media Dordrecht (outside the USA) 2016

Abstract In the southwestern United States (US), the Rio Grande chub (*Gila pandora)* is state-listed as a fish species of greatest conservation need and federally listed as sensitive due to habitat alterations and competition with non-native fishes. Characterizing genetic diversity, genetic population structure, and effective number of breeders will assist with conservation efforts by providing a baseline of genetic metrics. Genetic relatedness within and among *G. pandora* populations throughout New Mexico was characterized using 11 microsatellite loci among 15 populations in three drainage basins (Rio Grande, Pecos, Canadian). Observed heterozygosity (H_O) ranged from 0.71–0.87 and was similar to expected heterozygosity (0.75–0.87). Rio Ojo Caliente (Rio Grande) had the highest allelic richness ($A_R = 15.09$), while Upper Rio Bonito (Pecos) had the lowest allelic richness ($A_R = 6.75$). Genetic differentiation existed among all populations with the lowest genetic variation occurring within the Pecos drainage. STRUCTURE analysis revealed seven genetic clusters. Populations of *G. pandora* within the upper Rio Grande drainage (Rio Ojo Caliente, Rio Vallecitos, Rio Pueblo de Taos) had high levels of admixture with Q-values ranging from 0.30–0.50. In contrast, populations within the Pecos drainage (Pecos River and Upper Rio Bonito) had low levels of admixture (Q = 0.94 and 0.87, respectively). Estimates of effective number of breeders (N_b) varied from 6.1 (Pecos: Upper Rio Bonito) to 109.7 (Rio Grande: Rio Peñasco) indicating that populations in the Pecos drainage are at risk of extirpation. In the event that management actions are deemed necessary to preserve or increase genetic diversity of *G. pandora*, consideration must be given as to which populations are selected for translocation.

Keywords Gila pandora · Southwestern United States · Native Fish · Arid lands · Connectivity

Electronic supplementary material The online version of this article (doi:10.1007/s10592-016-0845-2) contains supplementary material, which is available to authorized users.

✉ Colleen A. Caldwell
ccaldwel@nmsu.edu

[1] Department of Fish, Wildlife and Conservation Ecology, New Mexico State University, 2980 South Espiña Street, Las Cruces, NM 88003, USA

[2] Southwestern Native Aquatic Resources and Recovery Center, U.S. Fish and Wildlife Service, 7116 Hatchery Road, Dexter, NM 88230, USA

[3] New Mexico Cooperative Fish and Wildlife Research Unit, U.S. Geological Survey, 2980 South Espiña Street, Las Cruces, NM 88003, USA

Introduction

The decline and eventual extirpation or extinction of native fishes can be attributed to degradation and fragmentation of habitat, introduction of non-native species, and conflict and competition for water. In the arid southwestern United States (US), population declines and risk of extinction can be exacerbated by drought and wildfire, which alter stream temperature, fish species composition, erosion patterns, water yield, and hydrologic processes (Bozek and Young 1994). Reduced flow, fragmented habitats, and the presence of non-native fishes have impeded re-colonization and persistence of native fishes shifting the fish community toward more thermally tolerant fish assemblages (Propst et al. 2008). Habitat connectivity within and among stream systems in the southwestern US is considered one of the more important factors for dispersal and re-colonization of extirpated fish populations (Fagan et al. 2002). As aquatic

habitat becomes fragmented, chances for re-colonization through dispersal processes become rare (Helfman et al. 2009). When populations become fragmented and isolated due to a lack of movement (Lande 1987), the strength of genetic drift increases due to a lack of gene flow. Depending on the demography of the population, genetic drift increases the rate at which genetic variation is lost and increases the risk of extinction (Lande 1988). This risk is offset by increasing or maintaining population connectivity to facilitate gene flow, which is essential for the long-term viability of a species (Segelbacher et al. 2010). Thus, understanding changes between historic and contemporary connectivity and gene flow is important for management and conservation of species to minimize the risk of extinction. As such, proactive planning and conservation goals should include genetic information to target species and community assemblages within ecological systems prior to landscape level alterations.

Of conservation interest in the southwestern US (southern Colorado and New Mexico) is a native fish assemblage that includes Rio Grande cutthroat trout (*Oncorhynchus clarkii virginalis*), Rio Grande sucker (*Catostomus plebeius*), and Rio Grande chub (*Gila pandora*) (Calamusso and Rinne 1996). Concerted management effort among state and federal agencies precluded listing of *O. c. virginalis* under the Endangered Species Act (U.S. Fish and Wildlife Service 2014). Similar efforts are underway with *C. plebeius*, in which a genetic analysis across the species range has identified factors that limit its distribution (McPhee et al. 2008). However, less is known about the distribution and genetic diversity throughout populations of *G. pandora*. The species is restricted to less than 25 % of its historical range (Rees and Miller 2005). Threats to *G. pandora* include introduction and range expansion of non-native fishes and overall degradation of habitat resulting in low recruitment (Calamusso 2005; Rees and Miller 2005). The distribution of *G. pandora* includes the Rio Grande drainage basin of southern Colorado, New Mexico, and Texas with the majority of the historical range in New Mexico (Biota Information System of New Mexico 2009). A single population resides as far north as Hot Creek in Conejos County (37°45′), Colorado, and as far south as Little Agua Creek in Jeff Davis County (30°45′), Texas (Sublette et al. 1990), neither of which were sampled in the current study. The status of the Texas population is uncertain, but presumed to be extant. Douglas and Douglas (2003, 2006) examined mitochondrial DNA diversity of *G. pandora* and reported between 1 and 12 haplotypes throughout populations in the Rio Grande drainage of southern Colorado and northern New Mexico. In New Mexico, the distribution of *G. pandora* includes the tributaries within three drainage basins (Rio Grande, Pecos, Canadian). The species may have gained access to the Pecos River drainage through headwater exchanges during the Pleistocene (Smith and Miller 1986), but was recently listed as non-indigenous to the Canadian drainage basin (Fuller 2015). The State of New Mexico listed *G. pandora* as a species of greatest conservation need (New Mexico Department of Game and Fish 2006) and the US Forest Service listed the species as sensitive in New Mexico (U.S. Forest Service 2013).

At present, the genetic diversity and population structure of *G. pandora* are unknown in New Mexico. An understanding of genetic diversity, population structure, and effective population size within and among populations throughout New Mexico will be critical in the management of the species to avoid risk of extinction. Conservation goals should include identifying areas that retain genetic diversity with the potential for long-term viability (Neely et al. 2001); and, should the need arise, these goals would prioritize which populations could serve as source populations for translocation in New Mexico. To aid in restoration and recovery efforts of *G. pandora*, we examined genetic diversity, population structure, and effective number of breeders throughout the species range among three drainage basins in New Mexico.

Materials and methods

Study areas and sample collection

A comprehensive genetic assessment for only New Mexico populations of *G. pandora* was considered for this study. In 2010, 15 populations were non-randomly selected from the Rio Grande, Pecos, and Canadian drainages (Fig. 1). Populations (Table 1) were identified throughout New Mexico from prior field studies (Calamusso 1992) and agency records (personal communications with G. Gustina, US Bureau of Land Management; R. Hansen and E. Frey, New Mexico Department of Game and Fish). Using a backpack electrofisher (Smith-Root LR-24), fish were collected from slow moving runs, pools, and undercut banks. Non-lethal tissue samples were collected on site by clipping the anal fin and placing the tissue into a 2.0 ml Nalgene™ cryogenic tube filled with 95 % ethanol for preservation (McPhee and Turner 2004). Fish were released at the capture location immediately after tissue collection. Populations in the Rio Hondo and Rio Ruidoso of the Pecos drainage are presumed extirpated (personal observations of the authors), thus the genetic structure within these watersheds could not be examined.

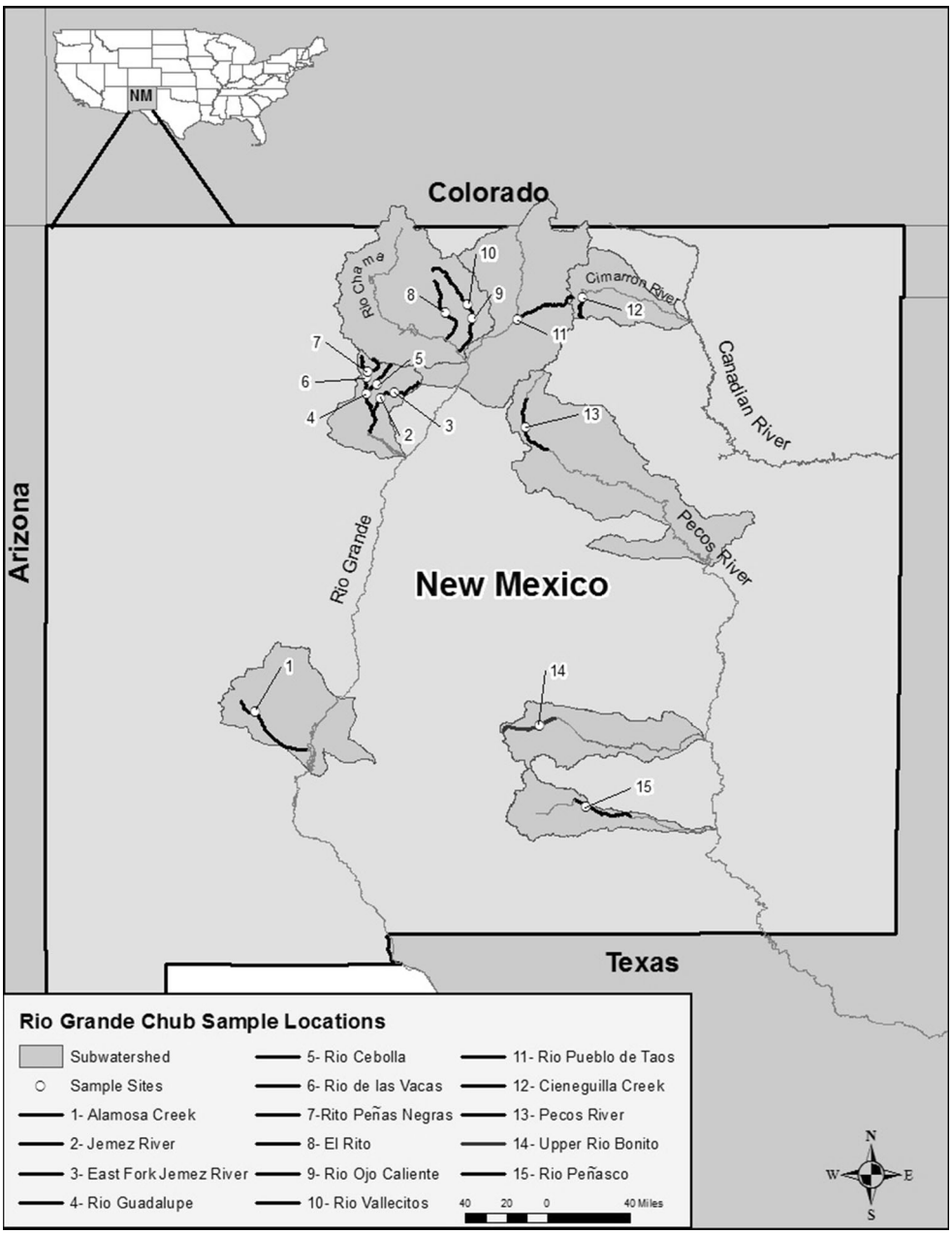

Fig. 1 Sampling locations of 15 populations of *G. pandora* within watersheds *dark gray shaded areas*. Populations numbers, drainage areas, and watersheds are listed in Table 1

DNA isolation and screening of microsatellite markers

Total genomic DNA was extracted from fin clips using a DNeasy Blood and Tissue Kit (Qiagen 2006). Eleven microsatellite loci were screened (Table S1) from microsatellites developed for four cyprinid species: bony-tail (*Gila elegans*; Keeler-Foster et al. 2004), Sacramento splittail (*Pogonichthys macrolepidotus,* Baerwald and May 2004), Sacramento pikeminnow (*Ptychocheilus grandis,* Baerwald and May 2004), and tui chub (*G. bicolor*; Baerwald and May 2004). The mix for the polymerase chain reaction (PCR) consisted of 3.5 µl ddH_2O, 1× GeneAmp®10× PCR Buffer, 2.5 mM $MgCl_2$, 1.5 mM dNTPs, 0.875 U AmpliTaq Gold® DNA polymerase, 0.5 µl each of forward and reverse primers. All PCR primers were purchased from Applied Biosystems (Foster City, CA). Amplification protocol used a touchdown thermal profile (annealing temperature decreased by 0.2 °C at each cycle) on a GeneAmp PCR System 9700 thermal cycler. The profile started with a 9 min initial denaturing step at 95 °C (activation of AmpliTaq Gold® DNA polymerase), 33 cycles at 95 °C for 45 s, 56 °C for 1 min, and 72 °C for 1 min with a final extension of 30 min at 72 °C. PCR

Table 1 *Gila pandora* sample collection including drainage, watershed, population, land ownership

Drainage/watershed	(Map number) population	Land ownership	Total number	Collection date	UTM (E)	UTM (N)
Rio Grande						
Elephant Butte Reservoir	(1) Alamosa Creek	State	30	2/14/2011	261152	3717396
Jemez River	(2) Jemez River	USFS-SFNF	30	6/10/2010	347507	3961478
	(3) East Fork Jemez River	USFS-SFNF	30	8/18/2010	356527	3965771
	(4) Rio Guadalupe	USFS-SFNF	30	8/25/2010	338560	3965077
	(5) Rio Cebolla	USFS-SFNF	30	8/18/2010	345507	3971767
	(6) Rio de las Vacas	USFS-SFNF	30	7/15/2010	338464	3977630
	(7) Rito Peñas Negras	USFS-SFNF	30	7/15/2010	339664	3982014
Rio Chama	(8) El Rito	USFS-CNF	30	5/31/2010	389424	4027618
	(9) Rio Ojo Caliente	BLM	30	9/20/2010	406630	4022484
	(10) Rio Vallecitos	USFS-CNF	30	9/19 2010	403895	4033935
Upper Rio Grande	(11) Rio Pueblo de Taos	BLM	30	9/13/2010	434728	4021645
Canadian						
Cimarron River	(12) Cieneguilla Creek	State	30	5/23/2010	476304	4038428
Pecos						
Pecos Headwaters	(13) Pecos River	State	30	9/25/2010	439327	3937179
Rio Hondo	(14) Upper Rio Bonito	BLM	30	10/11/2010	446819	3703745
Rio Peñasco	(15) Rio Peñasco	Private	25	4/9/2011	477332	3640086

Total number of individuals collected, date collected, and UTM coordinates (Zone 13 N)

USFS United States Forest Service, *SFNF* Santa Fe National Forest, *CNF* Carson National Forest, *BLM* Bureau of Land Management, *State* State of New Mexico owned public land, *Private* privately owned land

products were processed using an ABI 3130xl genetic analyzer and GeneScan™ 500 LIZ® size standard (Applied Biosystems, Foster City, CA). GeneMapper 4.0 (Applied Biosystems, Foster City, CA) was used to score and edit genotypes.

Statistical analysis

Departures from Hardy–Weinberg (HW) expectations and genotypic equilibrium were tested using GENEPOP'007 (Rousset 2008). Deviations from HW due to stuttering, null alleles, and large allele dropout were tested using MICRO-CHECKER 2.2.3 (Van Oosterhout et al. 2004). Allelic richness (A_R) and inbreeding coefficient (F_{IS}) were calculated using FSTAT 2.9.3 (Goudet 2001). Allelic richness is the mean number of alleles per locus adjusted for sample size using rarefaction methods (Leberg 2002). Observed (H_O) and expected (H_E) heterozygosity, allele frequencies, and private alleles were estimated using GENALEX 6 (Peakall and Smouse 2006). Tests of both HW and genotypic disequilibrium were evaluated after sequential Bonferroni correction (Holm 1979; Rice 1989).

Partitioning of genetic variation among samples was assessed using a hierarchical analysis of molecular variance (AMOVA) in ARLEQUIN v3.5 (Excoffier et al. 1992, 2005) for three different levels (within populations, among populations within a basin, and among basins) using Weir and Cockerham (1984). Values for F_{ST} were compared using Wright's guidelines (Wright 1978) and F_{SC} was used to compare populations within a basin. A Bayesian clustering approach implemented in STRUCTURE v2.3.3 identified genetic clusters of *G. pandora* and assigned individuals to ancestry clusters (K) (Pritchard et al. 2000) using the admixture model. A 200,000 burn-in period was followed by 20 iterations of 500,000 Markov Chain Monte Carlo replicates per K (1–17). The maximum number of populations (K) was determined by using the largest ΔK value (Evanno et al. 2005) as implemented in STRUCTURE HARVESTER (Earl and von Holdt 2012). CLUMPAK (Kopelman et al. 2015) was used to create optimal alignment plots of the STRUCTURE results.

A Discriminate Analysis of Principal Components (DAPC) was performed to visualize genetic differentiation and the relationship among population using the R package (R DevelopmentCore Team 2015) adegenet (Jombart 2008; Jombart and Ahmed 2011). This multivariate approach does not require linkage or HW equilibrium assumptions about an underlying genetic model (Jombart et al. 2008). Populations for the analysis were determined by sampling location. Cross-validation was performed to determine the number of principal components to retain based on an accumulated variance explained by the eigenvalues of the

principal component analysis. A discriminant analysis of the retained principal components was then performed on the loadings to visualize between-group variance and within-group variance. In addition, the optimal number of genetic clusters that best described the data was explored using the function *find.clusters*. The function runs the K-means sequentially with increasing values of K and compares models using Bayesian Information Criterion and plots these against increasing values of K (see Fig. S1).

Estimation of effective numbers of breeders (N_b) utilized the single sample linkage-disequilibrium method proposed by Hill (1981). The modified method corrects for bias due to sample size (see England et al. 2006) and is implemented in the program NeESTIMATOR v2 (Do et al. 2014). Alleles with frequencies less than 0.02 ($P_{crit} = 0.02$) were removed when estimating N_b (Waples and Do 2008; Do et al. 2014).

Results

Eleven loci were used to genotype 445 individuals of *G. pandora* (Table S1). Analysis of HW demonstrated that four loci were statistically significant (*Gel*222, *Cyp*G26, *Cyp*G5, *Gel*257) indicating departure from HW due to the presence of null alleles; however, these four loci were not removed since HW departures were randomly distributed among populations indicating departures were at the population level and not at the locus level. A test for linkage disequilibrium revealed two loci (*Gel*323 and *Gel*257) deviated from random association. Results did not change after reanalyzing the data without *Gel*323 and *Gel*257; therefore, these loci were retained and 11 loci were used for final analyses.

For 11 loci, total number of alleles (N_A) per locus (Table S2) ranged from 2 to 27 across populations with an average number of alleles from as low as 7 in the Pecos drainage (Upper Rio Bonito) to as high as 16 in the Rio Grande drainage (Rio Ojo Caliente). Allelic richness was lower than the total number of alleles due to an adjustment for sample size from 30 to 25 individuals across all populations. Allelic richness was lowest in the Upper Rio Bonito ($A_R = 6.75$) within the Pecos drainage while the Rio Ojo Caliente within the Rio Grande drainage exhibited the highest allelic richness ($A_R = 15.09$). Observed heterozygosity varied from 0.708 (Upper Rio Bonito) to 0.872 (Rio Ojo Caliente) while H_E varied from 0.747 (Upper Rio Bonito) to 0.857 (Rio Pueblo de Taos). The inbreeding coefficient within populations (F_{IS}) across all loci ranged from −0.033 (East Fork Jemez River) to 0.070 (Upper Rio Bonito) (Table S2).

The AMOVA revealed moderate genetic differentiation among all 15 populations of *G. pandora* ($F_{ST} = 0.080$, $P < 0.001$; Table 2). Genetic differences among drainage basins were low with an $F_{CT} = 0.021$ ($P < 0.001$; Table 2). However, the main source of variation was from individuals within populations (91.9 %) and not among populations within a basin (5.9 %) or among basins (2.1 %). Population pairwise F_{ST} values varied from little genetic differentiation among populations within the Rio Grande drainage (Rio Guadalupe: Rio de las Vacas: Rito Peñas Negras) to high genetic differentiation between two populations in the Rio Grande drainage and Pecos drainage (Upper Rio Cebolla: Upper Rio Bonito, $F_{ST} = 0.171$, $P < 0.001$; Table 2). Overall, the Pecos River had the highest pairwise F_{ST} values. The STRUCTURE analysis and optimal K suggested seven genetic clusters (Fig. 2). The Upper Rio Bonito displayed the least amount of admixture compared to all other populations within the Pecos drainage. Populations of *G. pandora* within the upper Rio Grande drainage (Rio Ojo Caliente, Rio Vallecitos, Rio Pueblo de Taos) had high levels of admixture with Q-values ranging from 0.30 to 0.50. In contrast, populations within the Pecos drainage (Pecos River and Upper Rio Bonito) had low levels of admixture ($Q = 0.94$ and 0.87, respectively; Fig. 2).

A total of 150 principal components and 14 discriminant analysis axes were retained In the DAPC analysis. The proportion of conserved variance was 96 %. Similar to the STRUCTURE results, the plot of Bayesian Information Criterion against K suggested 7–10 clusters would be most descriptive of the *G. pandora* data (Fig. S1). The majority of the clusters overlapped, including Alamosa Creek and El Rito, which indicates a genetic link between the northern and southern tributaries in the Rio Grande (Fig. 3a). Upper Rio Bonito did not overlap with other clusters. Likewise Upper Rio Cebolla was a distinct cluster, however, its elliptical space trended towards other clusters represented by sample locations within the Jemez River (Fig. 3a). When four clusters from the southern drainage basins (Alamosa Creek, Upper Rio Bonito, Rio Peñasco, Pecos River) were removed from the DAPC analysis, the final groupings are consistent with the STRUCTURE analysis (Fig. 3b).

Effective number of breeders (N_b) were lowest for *G. pandora* in Upper Rio Bonito ($N_b = 6.1$) and highest in Rio Peñasco ($N_b = 109.7$), both within the Pecos drainage (Table 3). Within the Rio Grande drainage, estimates of N_b ranged from 19.1 (Rio Vallecitos) to 89.1 (Rio Pueblo de Taos). The upper bounds for confidence intervals were infinite in six of the 15 populations while all the lower bound estimates were positive.

Table 2 Pairwise F_{ST} values for 15 New Mexico populations of *G. pandora*

Rio Grande								
	1	2	3	4	5	6	7	8
1	*	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
2	0.080	*	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
3	0.082	0.027	*	0.0000	0.0000	0.0000	0.0000	0.0000
4	0.076	0.026	0.040	*	0.0000	0.0991	0.7748	0.0000
5	0.104	0.063	0.073	0.068	*	0.0000	0.0000	0.0000
6	0.075	0.036	0.038	**0.004**	0.064	*	0.03604	0.0000
7	0.080	0.025	0.037	**-0.002**	0.065	**0.007**	*	0.0000
8	0.095	0.070	0.063	0.068	0.088	0.059	0.068	*
9	0.069	0.031	0.026	0.031	0.053	0.019	0.034	0.040
10	0.071	0.050	0.037	0.045	0.078	0.031	0.051	0.057
11	0.077	0.035	0.043	0.039	0.050	0.029	0.037	0.060
13	0.074	0.053	0.057	0.046	0.073	0.036	0.040	0.065
14	0.146	0.138	0.152	0.147	0.171	0.141	0.143	0.164
15	0.131	0.083	0.097	0.098	0.108	0.090	0.094	0.095
12	0.081	0.055	0.052	0.052	0.081	0.046	0.055	0.055

Rio Grande				Pecos			Canadian
	9	10	11	13	14	15	12
1	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
2	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
3	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
4	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
5	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
6	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
7	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
8	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
9	*	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
10	0.013	*	0.0000	0.0000	0.0000	0.0000	0.0000
11	0.029	0.041	*	0.0000	0.0000	0.0000	0.0000
13	0.032	0.051	0.050	*	0.0000	0.0000	0.0000
14	0.127	0.156	0.145	0.099	*	0.0000	0.0000
15	0.058	0.100	0.077	0.070	0.165	*	0.0000
12	0.038	0.050	0.047	0.058	0.164	0.101	*

Bolded F_{ST} values represent non-significant comparisons after Bonferroni corrections ($\alpha < 0.0004$). Rio Grande Drainage: Alamosa Creek (1); East Fork Jemez River (2); Jemez River (3); Rio Guadalupe (4); Rio Cebolla (5); Rio de las Vacas (6); Rito Peñas Negras (7); El Rito (8); Rio Ojo Caliente (9); Rio Vallecitos (10); Rio Pueblo de Taos (11). Canadian Drainage: Cieneguilla Creek (12). Pecos River Drainage: Pecos River (13); Upper Rio Bonito (14); Rio Peñasco (15). Numbers above diagonal are P values, and numbers below the diagonal are F_{ST} values

Discussion

Our analysis revealed seven genetic clusters of *G. pandora* throughout the 15 sampling localities in New Mexico. Three of these genetic clusters represent unique isolated populations of *G. pandora* in southern New Mexico (Alamosa Creek, Rio Bonito, Rio Peñasco). This genetic differentiation is most likely attributed to the combined effects of isolation during the last glacial period (Schmidly 1977) and recent fragmentation preventing gene flow (Turner et al. 2015). *Gila pandora* is a member of an assemblage of Rio Grande fishes (*O. c. virginalis* + *C. plebeius* + *G. pandora*) that likely established residence in the southern portion of the Rio Grande in the late Pleistocene during the last glacial period (Wisconsin) when the region was cooler and mesic (Schmidly 1977). An evaluation of distribution patterns of co-occurring species can assist with understanding genetic data (De Luca et al.

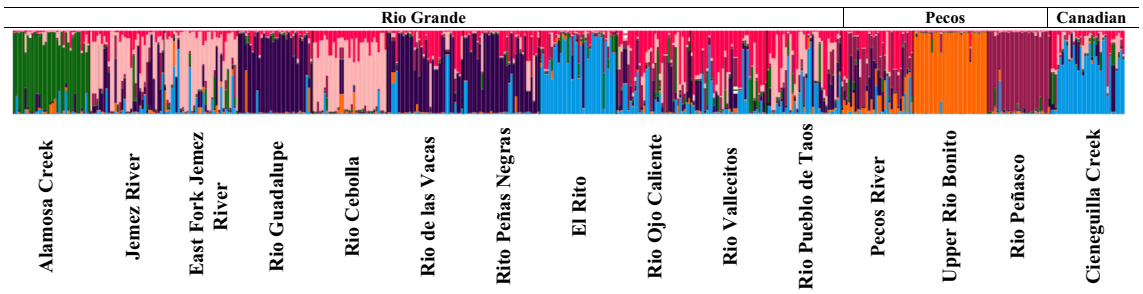

Fig. 2 *CLUMPAK* plot of *STRUCTURE* assignment results (K = 7)

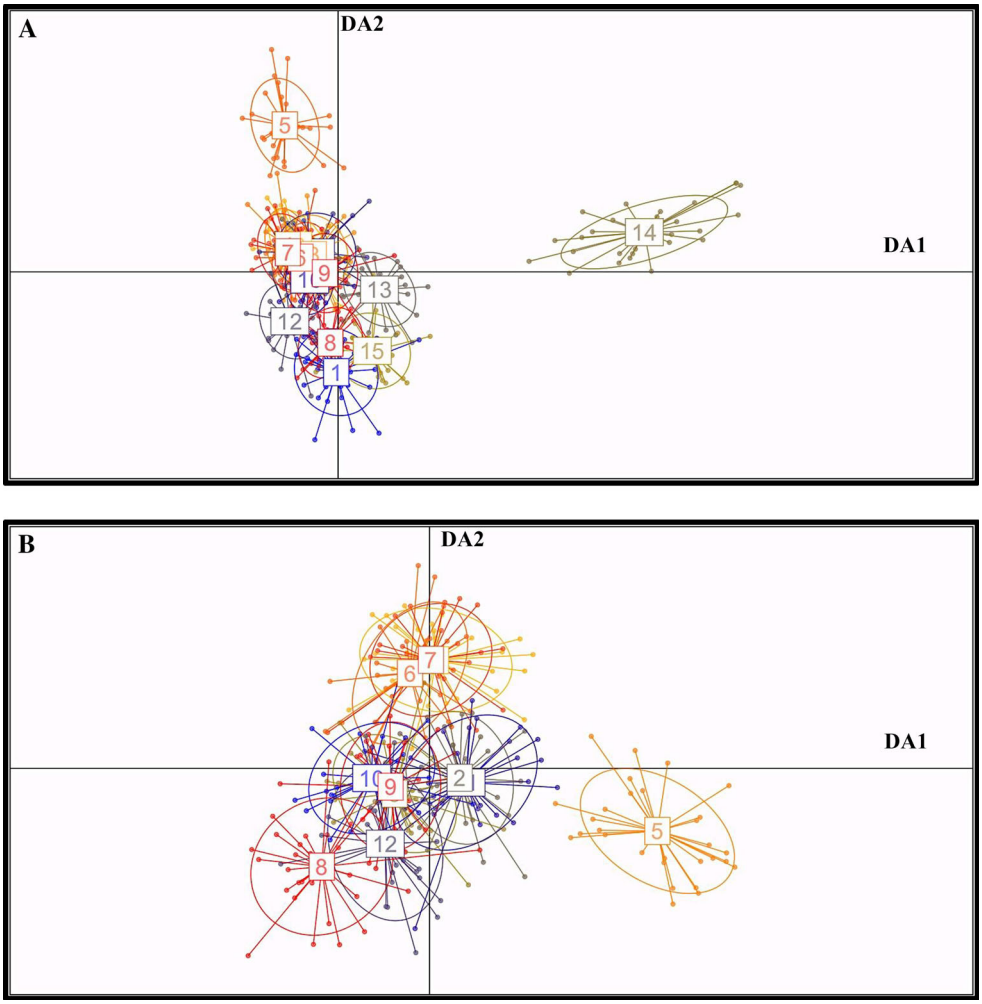

Fig. 3 Discriminate analysis of principal components (DAPC) scatterplots of *G. pandora* microsatellite data showing the first two principal components of the DAPC. Sampling locality was used as a prior. *Dots* represent individual fish and *ellipses* represent different sampling localities. **a** Plot of all 15 sample locations. **b** Plot of the eleven northern sample locations with southern sample locations 1, 14, and 15 removed, and sample location 13 from the Pecos drainage removed. The number of each sample locality is referenced in Fig. 1

2014); likewise, just as multiple independent genetic markers can reinforce observed patterns, comparison of genetic information among co-occurring species can strengthen genetic patterns (Avise et al. 1987). To this end, *G. pandora* is presumed to be native to Alamosa Creek because native populations of *C. plebeius* and *O. c. virginalis* (Rio Grande native fish assemblage) co-occur throughout these southern tributaries in the Rio Grande drainage. Similarly, McPhee et al. (2008) found that *C. plebeius* in Palomas Creek (an adjacent southwestern tributary to the lower Rio Grande) was genetically similar to *C. plebeius* populations throughout the northern Rio Grande drainage. In addition, *O. c. virginalis* was documented in a small tributary to the Rio Grande (Las Animas Creek) and represents the most southern distribution of the subspecies and that of all cutthroat trout (Behnke 1992). Thus, these documented historical distributions and genetic relatedness of the Rio Grande fish assemblage indicates a

Table 3 Effective number of *G. pandora* as breeders (N_b)

Drainage	Population	N_b	95 % CI
Rio Grande	Alamosa Creek	37.1	15.8–428.1
	East Fork Jemez River	43.3	20.3–286.8
	Jemez River	41.5	20.4–183.0
	Rio Guadalupe	63.1	24.9–∞
	Rio Cebolla	34.5	17.0–135.1
	Rio de las Vacas	42.1	21.0–177.7
	Rito Peñas Negras	50.8	21.7–3003.9
	El Rito	73.3	30.0–∞
	Rio Ojo Caliente	65.3	27.7–∞
	Rio Vallecitos	19.1	11.5–36.4
	Rio Pueblo de Taos	89.1	31.9–∞
Canadian	Cieneguilla Creek	49.8	19.5–∞
Pecos	Pecos River	64.7	29.5–1251.2
	Upper Rio Bonito	6.1	3.1–9.8
	Rio Peñasco	109.7	42.0–∞
	Harmonic Mean	32.4	

Estimates were calculated using the linkage-disequilibrium method as implemented in NeEstimator2 ($P_{CRIT} = 0.02$), 95 % CI from the Jackknife method

∞ represents infinity

historical connection between the southern and northern *G. pandora* populations.

Gila pandora is presumably native to the southern Pecos tributaries. Sublette (1975) described *G. pandora* in the Rio Ruidoso, Rio Felix, and Rio Peñasco (see map in Minckley 1980). The species presumably established populations throughout southern New Mexico during the Wisconsin. As the region became warmer and drier, *G. pandora* was relegated to the cooler tributaries of the Rio Hondo, Rio Bonito, and Rio Peñasco within the Pecos drainage. Although data are not available from McPhee et al. (2008) for cross species comparison, Sublette (1975) documented *C. plebeius* in the Rio Bonito and Behnke (1992) documented a resident population of *O. c. virginalis* in the Rio Bonito. *Gila pandora* in the Rio Bonito represents a unique population with low genetic diversity, a single private allele, a critically low estimate of effective number of breeders, and few low-frequency alleles. The reduced genetic diversity of the population in the Rio Bonito is likely due to drought and multiple water diversions that began in the early twentieth century. A major water diversion was created with the installation of a dam to form Bonito Lake in 1931 that was used by the Southern Pacific Railroad for steam engines (URH-WRAS 2004). Presumably, this diversion prevented movement of *G. pandora* among the tributaries and thereby increasing loss of genetic diversity through genetic drift (Haag et al. 2010).

Unlike the southern-most populations of *G. pandora*, population structure throughout the northern Rio Grande drainage (Jemez River, Rio Chama, Upper Rio Grande) revealed gene flow. Within the Jemez River watershed, populations from Rio Guadalupe, Rio de las Vacas, and Rito Peñas Negras exhibited very low F_{ST} values indicating connectivity and gene flow. *Gila pandora* from the Rio Cebolla, which flows into the Rio Guadalupe just below the confluence of the Rio de las Vacas and Rio Guadalupe, were genetically similar to both the Jemez River and East Fork Jemez River but not the Rio Guadalupe. We surmise that *G. pandora* reached the Rio Cebolla via stream capture or physical translocation from the Jemez River. The scenario of stream capture is plausible given that the upper portion of Rio Cebolla (ca. 2582 m elevation) runs parallel to San Antonio Creek (ca. 2552 m elevation) which is a tributary to the Jemez River. San Antonio Creek was once a lake (40 km^2) until the late Pleistocene. The Rio Cebolla population may have originated from San Antonio lake which had lacustrine deposits as high as 2650 m elevation and post-lake stream terrace at least 2700 m in elevation (Reneau et al. 2007). Thus, the low allelic diversity of the Rio Cebolla population could be linked to a founder event from stream capture between San Antonio Creek and the Rio Cebolla. If the Rio Cebolla population originated from stream capture, a biological or environmental barrier to fish movement into the Rio Guadalupe would have prevented gene flow between the two populations (gene flow between these two populations was not evident). Low allelic diversity of the Rio Cebolla population may be exacerbated by contemporary habitat loss because the stream currently contains little pool habitat and overall poor water quality (Simino and Bassett 2003).

Gila pandora populations within the Rio Chama and upper Rio Grande watersheds exhibited the overall greatest genetic diversity. With the exception of El Rito, tributaries within these watersheds were not dominated by a single genetic cluster. Rio Ojo Caliente, Rio Vallecitos, and Rio Pueblo de Taos contained ancestry from multiple genetic clusters indicating minimal drift and lineage sorting. With the exception of Rio Vallecitos ($N_b = 19.1$), estimated effective number of breeders for the two most northern watersheds were the highest (N_b range 65.3–89.1). The low N_b of Rio Vallecitos caused an increase in linkage disequilibrium as four locus pairs were out of equilibrium. These same loci were in equilibrium in Rio Chama and other upper Rio Grande populations. However, *G. pandora* in the Rio Vallecitos, tributary to the Rio Chama, may have linkage disequilibrium due to a bottleneck. The Rio Vallecitos is impaired due to high turbidity, high levels of aluminum, and high temperatures (RC-WRAS 2005). In contrast, neither Rio Ojo Caliente nor El Rito experience water quality impairment.

Gila pandora from Cieneguilla Creek, a tributary within the Canadian drainage basin in eastern New Mexico,

shared a unique genetic cluster with the population in El Rito (a tributary in the upper Rio Grande basin). Historical accounts of *G. pandora* in the Canadian River basin are few and often lack details. Sublette (1975) reported *G. pandora* in the Mora River, a tributary of the Canadian River; however, collection details were not provided. Minckley (1980) and Rees and Miller (2005) indicated that *G. pandora* may have been introduced into tributaries of the Canadian River, but again, details or citations were not provided. Evidence of distributions from the other two species of the native fish assemblage (*O. c. virginalis* and *C. plebeius*) is either conflicting or lacking (Pritchard et al. 2008).

The effective number of breeders (N_b) estimates for *G. pandora* were low, but in some cases similar to *G. nigra* (Headwater Chub; $N_{eD} = 61.0–112.0$) from the Gila River in New Mexico (Pilger et al. 2015) and *G. nigrescens* (Chihuahua Chub; $N_{eD} = 54.4–226.0$) from the Mimbres River, also in New Mexico (Osborne et al. 2012). In species with overlapping generations, these raw N_b estimates are likely biased downward due to the Wahlund effect with the magnitude of the bias affected by the number of cohorts within the sample (Waples 2005; Waples et al. 2014; see also Robinson and Moyer 2013). This bias could be corrected and used to estimate generational effective size (N_e) using several life history traits including adult lifespan and age at maturity (Waples et al. 2014); however, values of these traits are unknown for *G. pandora*. Relating the current N_b estimates to the N_e 50–500 rule (see Frankham et al. 2013; Jamieson and Allendorf 2012, 2013) is unwarranted at this time because $N_b \neq N_e$ (see Waples et al. 2014). That being said, an N_b of 6.1 (Upper Rio Bonito) is likely below an $N_e = 50$. This population is in danger of extirpation due to increased inbreeding (Palstra and Ruzzante 2008).

Conservation implications

Habitat loss was listed as the leading cause of decline in *G. pandora* (WildEarth Guardians 2013). Habitat loss can fragment and thus reduce genetic exchange among populations as well as increase susceptibility to extirpation from catastrophic events. We characterized genetic variation and population structure throughout the extant range of *G. pandora* in New Mexico to aid as reference points in conservation planning and recovery. We recommend incorporating genetic monitoring into conservation planning of *G. pandora* to determine how physical barriers of connectivity as well as biological barriers of non-native fishes affect genetic variation and number of breeders. Conservation planning should also take into consideration the impact that a changing climate will have on biological diversity (Comte et al. 2013). For example, Zeigler et al. (2012) reported an increase in air temperature (0.29 °C decade^{-1}), earlier onset of snowmelt (2.3 days decade^{-1}), and a decrease in snow-water equivalent (5.3 % decade^{-1}) throughout southern Colorado and New Mexico. Not only will these climate patterns affect the distribution and continued persistence of cool-water fishes such as *G. pandora*, but also such conditions increase the frequency and intensity of wildfires (Rieman and Isaak 2010). An example was the 2012 Whitewater-Baldy fire in the Gila National Forest of New Mexico where approximately 121,410 ha burned (Tillery et al. 2012). The fire eliminated several populations of the federally endangered Gila trout (*O. gilae*). Prior knowledge of genetic diversity resulted in a multi-agency response to rescue relict populations for preservation of extant genetic diversity. In the event that populations of *G. pandora* are subject to extirpation, then populations can be targeted for translocations to preserve genetic diversity. Proactive planning such as this would guide rescue and conservation efforts of *G. pandora* to ensure continued persistence in an arid landscape.

Acknowledgments Support was provided by the United States (US) Forest Service—Santa Fe National Forest (C. Cook) (IAG 09-IA-11031000-008), the US Bureau of Land Management (G. Gustina) (No. L09PG00403), and New Mexico Department of Game and Fish—Share with Wildlife Program. Field support was provided by R. Hansen of New Mexico Department of Game and Fish, M. Zeigler and S. Hall of New Mexico State University, Department of Fish, Wildlife and Conservation Ecology. Laboratory support was provided by R. Martin and M. Robinson, US Southwestern Aquatic Resources and Recovery Center. Additional support was provided by the Department of Fish, Wildlife and Conservation Ecology at New Mexico State University. This manuscript was substantially improved by comments from M. McPhee, R. Martin, M. Robinson, and T. Diver. Field collections were allowed under New Mexico Department of Game and Fish Authorization for Taking Protected Wildlife for Scientific and Educational Purposes Permit 3033 and New Mexico State University Institutional Animal Care and Use Committee Protocol 2011-003. Any use of trade, firm, or product names is for descriptive purposes only and does not imply endorsement by the United States Government.

References

Avise JC, Arnold J, Ball RM, Bermingham E, Lamb T, Neigel JE, Reeb CA, Saunders NC (1987) Intraspecific phylogeography: the mitochondrial DNA bridge between population genetics and systematics. Annu Rev Ecol Syst 18:489–522

Baerwald MR, May B (2004) Characterization of microsatellite loci for five members of the minnow family Cyprinidae found in the Sacramento-San Joaquin delta and its tributaries. Mol Ecol Notes 4:385–390

Behnke RJ (1992) Native trout of western North America. American Fisheries Society Monograph, No. 6. American Fisheries Society, Bethesda

Biota Information System of New Mexico (2009) Species: Rio Grande Chub (*Gila pandora*). Species Id 010140. 25 http://www.bison-m.org/booklet.aspx?id=010140. Accessed 6 Jan 2010

Bozek MA, Young MK (1994) Fish mortality resulting from delayed effects of fire in the greater yellowstone ecosystem. Great Basin Nat 54:91–95

Calamusso B (1992) Current distribution of *Catostomus plebeius* and *Gila pandora* on the Carson National Forest, New Mexico with preliminary comment on habitat preferences. Proc Desert Fishes Counc 24:63–64

Calamusso B (2005) Fishes of the southwestern grasslands: ecology, conservation, and management. In: Finch DM (ed) Assessment of grassland ecosystem conditions in the Southwestern United States: Wildlife and Fish. U.S. Forest Service General Technical Report RMRS-GTR-135, pp 141–168

Calamusso B, Rinne JN (1996) Distribution of Rio Grande cutthroat trout and its co-occurrence with the Rio Grande sucker and Rio Grande chub on the Carson and Santa Fe National Forest. U.S. Forest Service General Technical Report RMRS-GTR-272, pp 157–167

Comte L, Buisson L, Daufresne M, Grenouillet G (2013) Climate-induced changes in the distribution of freshwater fish: observed and predicted trends. Freshwater Biol 58:625–639

De Luca D, Catanese G, Procaccini G, Fiorito G (2014) An integration of historical records and genetic data to the assessment of global distribution and population structure in *Octopus vulgaris*. Front Ecol Evol 2:55

Do C, Waples RS, Peel D, Macbeth GM, Tillett BJ, Ovenden JR (2014) NeEstimator v2: re-implementation of software for the estimation of contemporary effective population size (*Ne*) from genetic data. Mol Ecol Resour 14:209–214

Douglas MR, Douglas ME (2003) Rio Grande chub genetic assessment final report. Colorado Division of Wildlife Project 5-30479. Department of Fishery and Wildlife Biology, Colorado State University, Fort Collins

Douglas MR, Douglas ME (2006) Rio Grande chub genetic assessment 3. Colorado Division of Wildlife Project 5-30479. Department of Fishery and Wildlife Biology, Colorado State University, Fort Collins

Earl DA, von Holdt BM (2012) STRUCTURE HARVESTER: a website and program for visualizing STRUCTURE output and implementing the Evanno method. Conserv Genet Resour 4:359–361

England PR, Cornuet JM, Berthier P, Tallmon DA, Luikart G (2006) Estimating effective population size from linkage disequilibrium: severe bias in small samples. Conserv Genet 7:303–308

Evanno G, Regnaut S, Goudet J (2005) Detecting the number of clusters of individuals using the software STRUCTURE: a simulation study. Mol Ecol 14:2611–2620

Excoffier L, Smouse PE, Quattro JM (1992) Analysis of molecular variance inferred from metric distance among DNA haplotypes: application to human mitochondrial DNA restriction data. Genetics 131:479–491

Excoffier L, Laval G, Schneider S (2005) Arlequin (version 3.0): an integrated software package for population genetics data analysis. Evol Bioinform Online 1:47–50

Fagan WF, Unmack PJ, Burgess C, Minckley WL (2002) Rarity, fragmentation, and extinction risk in desert fishes. Ecology 83:3250–3256

Frankham R, Brook BW, Bradshaw CJA, Traill LW, Spielman D (2013) 50/500 rule and minimum viable populations: response to Jamieson and Allendorf. Trends Ecol Evol 28:187–188

Fuller P (2015) *Gila pandora*. USGS nonindigenous aquatic species database, Gainesville. http://nas.er.usgs.gov/queries/FactSheet.aspx?speciesID=541. Revision Date 26 June 2000

Goudet J (2001) FSTAT, a program to estimate and test gene diversities and fixation indices. Version 2.9.3. www.unil.ch/izea/softwares/fstat.html. Accessed 7 Mar 2012

Haag T, Santos AS, Sana DA, Morato RG, Cullen L Jr, Crawshaw PG Jr, De Angelo C, Di Bitetti MS, Salazano FM, Eizirik E (2010) The effect of habitat fragmentation on the genetic structure of a top predator: loss of diversity and high differentiation among remnant populations of Atlantic forest jaguars (*Panthera onca*). Mol Ecol 19:4906–4921

Helfman GS, Collette BB, Facey DE, Bowen BW (2009) The diversity of fishes: biology, evolution, and ecology. Wiley-Blackwell, Chichester

Hill WG (1981) Estimation of effective population size from data on linkage disequilibrium. Genet Res 38:209–216

Holm S (1979) A simple sequentially rejective multiple test procedure. Scand J Stat 6:65–70

Jamieson IG, Allendorf FW (2012) How does the 50/500 rule apply to MVPs? Trends Ecol Evol 27:578–584

Jamieson IG, Allendorf FW (2013) A school of red herring: reply to Frankham et al. Trends Ecol Evol 28:188–189

Jombart T (2008) Adegenet: an R package for the multivariate analysis of genetic markers. Bioinformatics 27:3070–3071

Jombart T, Ahmed I (2011) Adegenet 1.3-1: new tools for the analysis of genome-wide SNP data. Bioinformatics 24:1403–1405

Jombart T, Devillard S, Dufour AB, Pontier D (2008) Revealing cryptic spatial patterns in genetic variability by a new multivariate method. Heredity 101:92–103

Keeler-Foster CL, Spies IB, Bondu-Hawkins V, Bentzen P (2004) Development of microsatellite markers in bonytail (*Gila elegans*) with cross-species amplification in humpback chub (*Gila cypha*). Mol Ecol Notes 4:23–25

Kopelman NM, Mayzel J, Jakobsson M, Rosenberg NA, Mayrose I (2015) CLUMPAK: a program for identifying clustering modes and packaging population structure inferences across K. Mol Ecol Resour 15:1179–1191

Lande R (1987) Extinction thresholds in demographic models of territorial populations. Am Nat 130:624–635

Lande R (1988) Genetics and demography in biological conservation. Science 241:1455–1460

Leberg PL (2002) Estimating allelic richness: effects of sample size and bottlenecks. Mol Ecol 11:2445–2449

McPhee MV, Turner TF (2004) No genetic evidence for hybridization between the Rio Grande sucker, *Catostomus plebeius,* and the introduced white sucker, *Catostomus commersoni*, in the Rio Grande. Environ Biol Fishes 71:85–93

McPhee MV, Osborne MJ, Turner TF (2008) Genetic diversity, population structure, and demographic history of the Rio Grande sucker, *Catostomus* (*Pantosteus*) *plebeius*, in New Mexico. Copeia 2008:191–199

Minckley WL (1980) *Gila pandora* (Cope) Rio Grande chub. In: Lee DS (ed) Atlas of North American freshwater fishes. North Carolina Biological Survey, Raleigh, p 170

Neely B, Comer P, Moritz C, Lammert M, Rondeau R, Pague C, Bell G, Copeland H, Humke J, Spackman S, Schulz T, Theobald D, Valutis L (2001) Southern Rocky Mountains: an ecoregional assessment and conservation blueprint. Prepared by The Nature Conservancy with support from the U.S. Forest Service, Rocky Mountain Region, Colorado Division of Wildlife, and U.S. Bureau of Land Management

New Mexico Department of Game and Fish (2006) Comprehensive wildlife conservation strategy for New Mexico. New Mexico Department of Game and Fish, Santa Fe

Osborne M, Sharp A, Monzingo J, Propst DL, Turner TF (2012) Genetic analysis suggests high conservation value of peripheral populations of Chihuahua chub (*Gila nigrescens*). Conserv Genet 13:1317–1328

Palstra FP, Ruzzante DE (2008) Genetic estimates of contemporary effective population size: what can they tell us about the

importance of genetic stochasticity for wild population persistence? Mol Ecol 17:3428–3447

Peakall R, Smouse PE (2006) GENALEX 6: genetic analysis in Excel. Population genetic software for teaching and research. Mol Ecol Notes 6:288–295

Pilger TJ, Gido KB, Propst DL, Whitney JE, Turner TF (2015) Comparative conservation genetics of protected endemic fishes in an arid-land riverscape. Conserv Genet 16:875–888

Pritchard JK, Stephens M, Donnelly P (2000) Inference of population structure using multilocus genotype data. Genetics 155:945–959

Pritchard VL, Metcalf JL, Jones K, Martin AP, Cowley DE (2008) Population structure and genetic management of Rio Grande cutthroat trout (*Oncorhynchus clarkii virginalis*). Conserv Genet 10:1209–1221

Propst DL, Gido KB, Stefferud JA (2008) Natural flow regimes, nonnative fishes, and native fish persistence in arid-land river systems. Ecol Appl 18:1236–1252

QIAGEN (2006) Dneasy blood and tissue handbook. QIAGEN Group. http://www1.qiagen.com/literature/Default.aspx?Term=Dneasy&Language=EN&LiteratureType=1+2+3+4&ProductCategory=0. Accessed 11 Jan 2010

R Development Core Team (2015) R: a language and environment for statistical computing. R Foundation for Statistical Computing. https://www.R-project.org

RC-WRAS (2005) Rio Chama watershed restoration action strategy (WRAS). https://www.env.nm.gov/swqb/wps/WRAS/ChamaWRAS2005.pdf

Rees RJ, Miller WJ (2005) Rio Grande chub (*Gila pandora*): a technical conservation assessment. Prepared for the U.S. Department of Agriculture, Forest Service, Rocky Mountain Region, Species Conservation Project

Reneau SL, Drakos PG, Katzman D (2007) Post-resurgence lakes in the Valles Caldera, New Mexico. New Mexico Geological Society guidebook. In: 58th Field Conference, Geology of the Jemez Mountains Region II, pp 398–408

Rice WR (1989) Analyzing tables of statistical tests. Evolution 43:223–225

Rieman BE, Isaak DJ (2010) Climate change, aquatic ecosystems, and fishes in the Rocky Mountain West: implications and alternatives for management. General Technical Reports RMRS-GTR-250. Fort Collins, Colorado: U.S. Department of Agriculture, Forest Service, Rocky Mountain Research Station

Robinson JD, Moyer GR (2013) Linkage disequilibrium and effective population size when generations overlap. Evol Appl 6:290–302

Rousset F (2008) GENEPOP'007: a complete re-implementation of the GENEPOP software for Windows and Linux. Mol Ecol Resour 8:103–106

Schmidly DJ (1977) Factors governing the distribution of mammals in the Chihuahuan Desert region. In: Wauer RH, Riskind EH (eds). Transactions of the Symposium on the Biological Resources of the Chihuahuan Desert region, United States and Mexico. Proceedings and Transaction Series No. 3, U.S. Department of Interior, National Park Service

Segelbacher G, Cushman SA, Epperson BK, Fortin MJ, Francois O, Hardy OJ, Holderegger R, Taberlet P, Waits LP, Manel S (2010) Applications of landscape genetics in conservation biology: concepts and challenges. Conserv Genet 11:375–385

Simino J, Bassett M (2003) Rio Cebolla stream inventory report. United States Forest Service, Santa Fe National Forest, Santa Fe. http://www.fs.usda.gov/Internet/FSE_DOCUMENTS/fsbdev7_020815.pdf

Smith ML, Miller RR (1986) The evolution of the Rio Grande basin as inferred from its fish fauna. In: Hocutt CH, Wiley EO (eds) The zoogeography of North American freshwater fishes. Wiley, New York

Sublette JE (1975) A survey of the fishes of the Pecos, Canadian and Arkansas drainages in New Mexico. Report to New Mexico Department of Game and Fish, Santa Fe, 15 pp + appendices

Sublette JE, Hatch MD, Sublette M (1990) The fishes of New Mexico. University of New Mexico Press, Albuquerque

Tillery AC, Matherne AM, Verdin KL (2012) Estimated probability of postwildfire debris flows in the 2012 Whitewater-Baldy Fire burn area, southwestern New Mexico. U.S. Geological Survey Open-File Report 2012-1188

Turner TF, Osborne MJ, McPhee MV, Kruse CG (2015) High and dry: intermittent watersheds provide a test case for genetic response of desert fishes to climate change. Conserv Genet 16:399–410

URH-WRAS (2004) The Upper Rio Hondo Watershed Coalition—watershed restoration action strategy (WRAS) Upper Rio Hondo, 2nd Edn. http://www.nmenv.state.nm.us/swqb/wps/WRAS/Upper_Hondo_13060008_WRAS_Apr_2004.pdf

U.S. Fish and Wildlife Service (2014) Twelve-month finding on a petition to list Rio Grande cutthroat trout as an endangered or threatened species. Fed Regist 79:59140-59150. Accessed 1 Oct 2014)

U.S. Forest Service (2013) USFS R3 Regional Forester's Sensitive Species: animals. http://www.fs.usda.gov/Internet/FSE_DOCUMENTS/fsbdev3_021328.pdf

Van Oosterhout C, Hutchinson WF, Wills DPM, Shipley P (2004) Micro-Checker: software for identifying and correcting genotyping errors in microsatellite data. Mol Ecol Notes 4:535–538

Waples RS (2005) Genetic estimates of contemporary effective population size: to what time periods do the estimates apply? Mol Ecol 14:3335–3352

Waples RS, Do C (2008) LDNE: a program for estimating effective population size from data on linkage disequilibrium. Mol Ecol Resour 8:753–756

Waples RS, Antao T, Luikart G (2014) Effects of overlapping generations on linkage disequilibrium estimates of effective population size. Genetics 197:769–780

Weir BS, Cockerham CC (1984) Estimating F-statistics for the analysis of population structure. Evolution 3:1358–1370

WildEarth Guardians (2013) Petition to list the Rio Grande chub (*Gila pandora*) under the endangered species act. Petition submitted to the U.S. Secretary of the Interior acting through the U.S. Fish and Wildlife Service. Accessed 27 Sep 2013

Wright S (1978) Evolution and the genetics of population: variability within and among natural populations, vol 4. University of Chicago Press, Chicago

Zeigler MP, Todd A, Caldwell CA (2012) Evidence of recent climate change within the historic range of Rio Grande cutthroat trout: implications for management and future persistence. Trans Am Fish Soc 141:1045–1059